SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO2 AT1 - Capstone Infrastructure Sizing Workbook

Guided calibration workbook for VM sizing, virtualization decisions, snapshot planning and VM-container comparison

Assessment Metadata

Course Code	CSA15
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT1 - Virtualization Scenario Problem-Solving
Submission Type	Individual digital workbook based on the same capstone title used in CO1
Faculty	Dr. C. Rajesh Babu
Employee ID	SSETSCS-415
Submission Mode	Completed workbook and evidence repository link through Google Classroom

How This Workbook Works

- Read each guidance block and immediately fill the related response table.
- Use the same capstone title selected for CO1.
- Make reasonable assumptions when exact values are unknown. Good assumptions must be written clearly.
- The goal is not to guess a perfect industry configuration; the goal is to reason technically and justify the first infrastructure plan.
- The final decisions from this workbook will be used again in CO2 AT2 and CO2 AT3.

Technical Tip

A good infrastructure estimate is transparent. If a number is assumed, write why it is assumed. If a number is calculated, show the formula. If a service is selected, justify why it fits the workload.

Student and Capstone Details

Student Name	PAWAN KALYAN R
Register Number	1924241116
Class / Slot	Slot B
Capstone Title	Weather temperature forecasting using stacked RNNs
Capstone Product Type	Web app / Android app / iOS app / Hybrid app
CO1 Architecture Reference	Mention concept map / presentation title / GitHub link if available

Activity 1: Bring Forward CO1 Capstone Context

Start with the cloud product idea already used in CO1. The infrastructure plan must support that same application, not a new unrelated example.

Capstone Element	Student Entry
Primary users	Meteorologists, Researchers, Weather Analysts, General Users
User roles	Admin, Researcher, User
Main modules	Data Collection, Data Preprocessing, Stacked RNN Model, Temperature Prediction, Result Dashboard
Cloud services identified in CO1	Virtual Machine, Object Storage, Cloud Database
API/backend need	Python Flask REST API
Database need	MySQL
File upload/storage need	Weather CSV datasets
Analytics/AI/Python module	TensorFlow/Keras Stacked RNN

Activity 2: Workload Classification

Classify the capstone workload before selecting a VM. A form-based SaaS app and an AI analytics app should not receive the same infrastructure plan.

Workload Type	Typical Signals	Starting VM Guidance
Light web / form app	Login, CRUD, simple reports, low file upload	1-2 vCPU, 2-4 GB RAM
API backend	Authentication, REST APIs, moderate concurrent requests	2 vCPU, 4-8 GB RAM
Database-heavy app	Search, many records, frequent writes	2-4 vCPU, 8-16 GB RAM
File-heavy app	Images, PDFs, uploads, downloads	2 vCPU, 4-8 GB RAM + separate object storage
Analytics app	Dashboards, aggregation, batch reports	4 vCPU, 16 GB RAM
AI/Python module	Prediction, NLP, image/text processing	4-8 vCPU, 16-32 GB RAM; GPU optional
High traffic app	Large user base and peak demand	Multiple VMs or autoscaling group

Technical Tip

Start small for prototype, but do not hide growth. A valid answer can say: prototype uses one VM, pilot separates database, production uses load balancing and managed services.

Student Decision: My capstone workload type and reason

- **Workload Type:** AI/Python Analytics Application
- **Reason:** Uses deep learning (Stacked RNN) for temperature prediction and requires model training and inference.

Activity 3: User and Traffic Calibration

Use the following formulas to convert user assumptions into approximate traffic.

Metric	Formula	Example
urrent Active Users	Registered Users x Active User % x Peak Factor	500 x 10% x 2 = 100

ests per Minute	Concurrent Active Users x Actions per User per Minute	100 x 2 = 200
ests per Second	Requests per Minute / 60	200 / 60 = 3.33 RPS
Requests	Daily Active Users x Avg Actions per Day	200 x 30 = 6000

Input	Value	Reason
stered users expected	500	Prototype application
ve user percentage	10%	Moderate usage
k factor	2	Peak traffic
ons per minute	2	Prediction requests
ulated RPS	1.67 RPS	(500×10%×2×2)/60
y active users	200	Estimated
y requests	6000	Average usage

Activity 4: CPU Calibration

CPU sizing depends on request rate and processing complexity. Use the score below as a guided estimate.

CPU Driver	Score
se web/API application	1
thentication and role management	+1
oderate API traffic above 5 RPS	+1
tabase-heavy search or reporting	+1 to +2
ckground jobs or notifications	+1
alytics or dashboards	+2
Python inference inside server	+2 to +4

Total CPU Score	Recommended vCPU
1-2	1-2 vCPU
3-4	2 vCPU
5-6	4 vCPU
7-9	4-8 vCPU or separate analytics/AI service
10+	Scale out with multiple VMs or managed compute

CPU Driver	Score	Reason
se Web/API	1	Web interface
thentication	1	Login system

I Traffic	0	Below 5 RPS
tabase/Search	1	Weather data queries
ckground Jobs	1	Model scheduling
alytics Dashboard	2	Graph generation
/Python Module	3	Stacked RNN inference
U Driver	**Score**	**Reason**
se Web/API	1	Web interface

Activity 5: RAM Calibration

RAM must include operating system, runtime, application process, database/cache and a safety buffer.

RAM Formula

Estimated RAM = OS/runtime + app/API + database/cache + analytics/AI + 25% buffer. If database is managed separately, do not include full database memory in the app VM.

Component	Guidance	Student Estimate
OS + runtime	1-2 GB	2GB
Web/API service	1-2 GB	2GB
Small database	2-4 GB	4GB
Medium database/cache	4-8 GB	4GB
Analytics/reporting	4-8 GB extra	8GB
AI/Python module	4-16 GB extra	5GB
25% buffer	Subtotal x 0.25	
Final RAM	Round up to standard size	32GB

Activity 6: Storage Calibration

Storage must cover application files, database records, uploads, logs and backup/growth buffer.

Storage Item	Formula / Guidance	Student Estimate
OS and application	20-40 GB	
Database	Avg record KB x records/day x days x 1.3 / 1024 MB	30GB
File uploads	Avg file MB x files/day x days x 1.3	25GB
Logs	Requests/day x log KB x days / 1024 MB	30GB
Backup/growth buffer	Subtotal x 30%	10GB

Final storage	Round up to standard disk size	30GB

Technical Tip

Do not store large uploads permanently inside the VM disk when object storage is available. VM disk is for OS and application. Object storage is better for images, PDFs, media and documents.

Activity 7: Select the VM Plan

Plan	When It Fits	Student Choice
Single VM prototype	Small demo, simple app, low traffic	NO
Two-tier design	App VM + managed/separate database	YES
Three-tier design	Frontend, backend/API and database separated	NO
Containerized backend	Microservices, portable API, CI/CD	YES
Serverless function	Event-based task, notification, small automation	NO
Managed database/storage	Reliability, backup and scaling required	YES

Final VM Plan: selected plan, vCPU, RAM, storage and reason

Final VM Plan

- **Plan:** Two-tier Architecture
- **CPU:** 8 vCPU
- **RAM:** 32 GB
- **Storage:** 128 GB SSD
- **Reason:** Supports AI model training, REST API, and database efficiently.

Activity 8: Hypervisor and Virtualization Decision

Approach	Meaning	Best Use
Type 1 hypervisor	Runs directly on physical hardware	Production data center and cloud provider infrastructure
Type 2 hypervisor	Runs above an existing OS	Student lab, local testing, VirtualBox/VMware Workstation
Cloud-managed virtualization	Cloud provider hides hypervisor details	Most public cloud VM deployments
Container runtime	OS-level process isolation	Lightweight APIs, microservices, fast deployment

Context	Selected Approach	Justification
Local prototype/testing	Type-2 Hypervisor	VMware/VirtualBox
Cloud pilot deployment	Cloud-managed Virtualization	AWS/Azure/GCP VM

Production growth scenario	Type-1 Hypervisor	Better performance
Module suitable for container	Docker	Deploy Python backend

Activity 9: Snapshot and Clone Strategy

Snapshot protects a point in time. Clone creates a duplicate environment. Both are useful, but they are not the same.

Event	Action	Timing	Reason
Before Deployment	Snapshot	Every release	Rollback
Before DB Change	Snapshot	Before update	Data protection
Before OS Update	Snapshot	Monthly	Recovery
Before Review	Snapshot	Before demo	Stable version
Testing Environment	Clone	When required	Independent testing
Team Demo	Clone	Before presentation	Separate environment

Technical Tip
A snapshot is useful for rollback. A clone is useful for parallel testing. A clone should not be treated as the only backup.

Activity 10: VM vs Container Decision

Criteria	Virtual Machine	Container
Isolation	Strong OS-level isolation	Process-level isolation
Startup	Slower	Faster
Resource use	Higher	Lower
Best for	Full OS, legacy app, database VM	Microservices, APIs, portable services
Portability	Moderate	High
Student use	Clear infrastructure learning	Modern deployment learning
Capstone Module	**VM / Container / Managed Service**	**Reason**
Frontend	Container	Easy deployment
Backend/API	Container	Flask API
Database	Managed Service	Reliable backup
File storage	Managed Storage	Stores datasets
AI/Python module	VM	Requires more computing resources
Analytics/dashboard	Container	Lightweight
Notification/background jobs	Container	Scheduled processing

Activity 11: Final Recommendation

Write the final infrastructure recommendation in 8-12 technical lines. Include VM size, hypervisor/virtualization approach, storage plan, snapshot/clone plan and VM-container decision.

The proposed infrastructure uses a **Two-tier cloud architecture** for the Weather Temperature Forecasting application. The system will run on an **8 vCPU Virtual Machine with 32 GB RAM and 128 GB SSD storage**. A **Flask REST API** hosts the Stacked RNN prediction model, while **MySQL** stores weather datasets and prediction results. **Docker containers** are used for the frontend, backend, and dashboard services to simplify deployment. A **managed database and cloud object storage** improve scalability and reliability. **Cloud-managed virtualization** is selected for deployment, while **Type-2 virtualization** is used for local testing. Regular **snapshots** are created before deployments and updates, and **clones** are used for testing and demonstrations. This infrastructure provides good performance, scalability, and fault tolerance for AI-based weather forecasting.

Carry-Forward to CO2 AT2 and CO2 AT3

Output from CO2 AT1	How It Will Be Used Later
VM sizing values	Used as configuration target in CO2 AT2 practical evidence
Hypervisor/virtualization choice	Used to justify deployment environment
Snapshot/clone plan	Used as backup and testing evidence
VM/container comparison	Used in CO2 AT3 infrastructure design case
Final recommendation	Used as the capstone infrastructure baseline

GitHub Submission Checklist

- Completed workbook file.
- README with capstone title, sizing summary and final recommendation.
- Optional architecture diagram or table exported as image/PDF.
- Repository link submitted in Google Classroom.